5.9.3 SUMMARY

- The parts that make up a computer can be divided into three broad categories or subsystems: the central processing unit (CPU), the main memory, and the input/output subsystem.
- The central processing unit (CPU) performs operations on data. It has three parts: an arithmetic logic unit (ALU), a control unit, and a set of registers. The arithmetic logic unit (ALU) performs logic, shift, and arithmetic operations on data. Registers are fast stand-alone storage locations that hold data temporarily. The control unit controls the operation of each part of the CPU.
- Main memory is a collection of storage locations, each with a unique identifier called the *address*. Data is transferred to and from memory in groups of bits called *words*. The total number of uniquely identifiable locations in memory is called the *address space*. Two types of memory are available: random access memory (RAM) and read-only memory (ROM).
- The collection of devices referred to as the input/output (I/O) subsystem allows a computer to communicate with the outside world and to store programs and data even when the

When the power is off. Input/output devices can be divided into two broad categories: nonstorage and storage devices. Nonstorage devices allow the CPU/memory to communicate with the outside world. Storage devices can store large amounts of information to be retrieved at a later time. Storage devices are categorized as either magnetic or optical.

- ☐ The interconnection of the three subsystems of a computer plays an important role, because information needs to be exchanged between these subsystems. The CPU and memory are normally connected by three groups of connections, each called a *bus:* data bus, address bus, and control bus. Input/output devices are attached to the buses through an *input/output controller* or interface. Several kinds of controllers are in use. The

most common ones today are SCSI, FireWire, and USB.

- ☐ There are two methods of handling the addressing of I/O devices: isolated I/O and memory-mapped I/O. In the isolated I/O method, the instructions used to read/write to and from memory are different from the instructions used to read/write to and from input/output devices. In the memory-mapped I/O method, the CPU treats each register in the I/O controller as a word in memory.
- ☐ Today, general-purpose computers use a set of instructions called a *program* to process data. A computer executes the program to create output data from input data. Both the program and the data are stored in memory. The CPU uses repeating machine cycles to execute instructions in the program, one by one,

from beginning to end. A simplified cycle can consist of three phases: *fetch*, *decode*, and *execute*.

- ☐ Three methods have been devised for synchronization between I/O devices and the CPU: programmed I/O, interrupt-driven I/O, and direct memory access (DMA).
- ☐ The architecture and organization of computers have gone through many changes during recent decades. We can divide computer architecture into two broad categories: CISC (complex instruction set computers) and RISC (reduced instruction set computers).
- ☐ Modern computers use a technique called *pipelining* to improve their throughput. The idea is to allow the control unit to perform two or three phases simultaneously, which means that processing of the next instruction can start before the previous one is finished.
- ☐ Traditionally, a

- Traditionally, a computer had a single control unit, a single arithmetic logic unit, and a single memory unit. Parallel processing can improve throughput by using multiple instruction streams to handle multiple data streams.

5.10 PRACTICE SET

5.10.1 QUIZZES

A set of interactive quizzes for this chapter can be found on the book's website. It is strongly recommended that the student takes the quizzes to check his/her understanding of the materials before continuing with the practice set.

5.10.2 REVIEW QUESTIONS

Q5-1. What are the three subsystems that make up a computer?
Q5-2. What are the components of a CPU?
Q5-3. What is the function of the ALU?
Q5-4. What is the function of the control unit?
Q5-5. What is the function of main memory?
Q5-6. Defne RAM, ROM, SRAM, DRAM PROM, EPROM, and EEPROM.
Q5-7. What is the purpose of cache memory?
Q5-8. Describe the physical components of a magnetic disk.
Q5-9. How are the surfaces of a magnetic disk and magnetic tape organized?
Q5-10. Compare and contrast CD-R, CD-RW, and DVD.
Q5-11. Compare and contrast SCSI, FireWire and USB controllers.
Q5-12. Compare and contrast the two methods for handling the addressing of I/O devices.
Q5-13. Compare and contrast the three methods for handling the synchronization of the CPU with I/O devices.
Q5-14. Compare and contrast CISC architecture with RISC architecture.
Q5-15. Describe pipelining and its purpose.
Q5-16. Describe parallel processing and its purpose.

5.10.3 PROBLEMS

P5-1. A computer has 64 MB (megabytes) of memory. Each word is 4 bytes. How many bits are needed to address each single word in memory?
P5-2. How many bytes of memory are needed to store a full screen of data if the screen is made of 24 lines with 80 characters in each line? The system uses ASCII code, with each ASCII character stored as a byte.
P5-3. An imaginary computer has 16 data registers (R0 to R15), 1024 words in memory, and 16 different instructions (add, subtract, and so on). What is the minimum size of an add instruction in bits if a typical instruction uses the following

format: *add M R2*.

P5-4. If the computer in P5-3 uses the same size of word for data and instructions, what is the size of each data register?

P5-5. What is the size of the instruction register in the computer in P5-3?

P5-6. What is the size of the program counter in the computer in P5-3?

P5-7. What is the size of the data bus in the computer in P5-3?

P5-8. What is the size of the address bus in the computer in P5-3?

P5-9. What is the minimum size of the control bus in the computer in P5-3?

P5-10. A computer uses isolated I/O addressing. Its memory has 1024 words. If each controller has 16 registers, how many controllers can be accessed by this computer?

P5-11. A computer uses memory-mapped I/O addressing. The address bus uses ten lines (10 bits). If memory is made up of 1000 words, how many four-register controllers can be

accessed by the computer?

P5-12. Using the instruction set of the simple computer in Section 5.8, write the code for a program that performs the following calculation: D ← A + B + C, in which A, B, C, and D are integers in two's complement format. The user types the value of A, B, and, C, and the value of D is displayed on the monitor.

P5-13. Using the instruction set of the simple computer, write the code for a program that performs the following calculation:

B ← A + 3

A and 3 are integers in two's complement format. The user types the value of A and the value of B is displayed on the monitor. (Hint: use the increment instruction.)

P5-14. Using the

instruction set of the simple computer in Section 5.8, write the code for a program that performs the following calculation:

$B \leftarrow A - 2$

A and are integers in two's complement format. The user types the value of A and the value of B is displayed on the monitor. (Hint: use the decrement instruction.)

P5-15. Using the instruction set of the simple computer in Section 5.8, write the code for a program that adds *n* integers typed on the keyboard and displays their sum. You need first to type the value of *n*. (Hint: use decrement and jump instructions and repeat the addition *n* times.)

P5-16. Using the instruction set of the simple computer in Section 5.8, write the

code for a program that accepts two integers from the keyboard. If the first integer is 0, the program increment the second integer, and if the first integer is 1, the programs decrement the second integer. The first integer must be only 0 or 1 otherwise the program fails. The program displays the result of the increment or decrement.